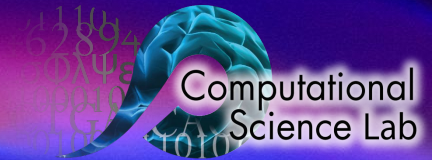


Swarmalators

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Joeri Fresen,
Margarita Petrova,
Leo Metasch



UNIVERSITEIT VAN AMSTERDAM

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What do they have in common?

Japanese Tree Frog



Drones



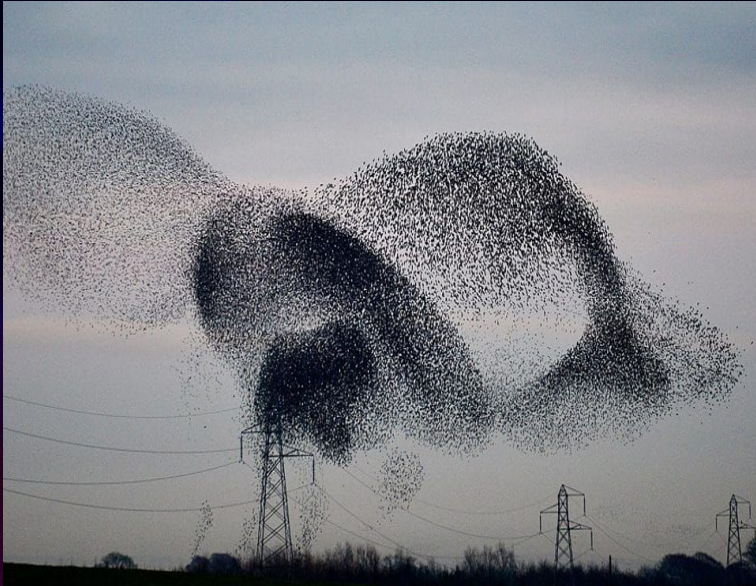
Fireflies



movement, internal phases and synchronisation $\Rightarrow$ collective behaviour from simple rules

Swarmalators

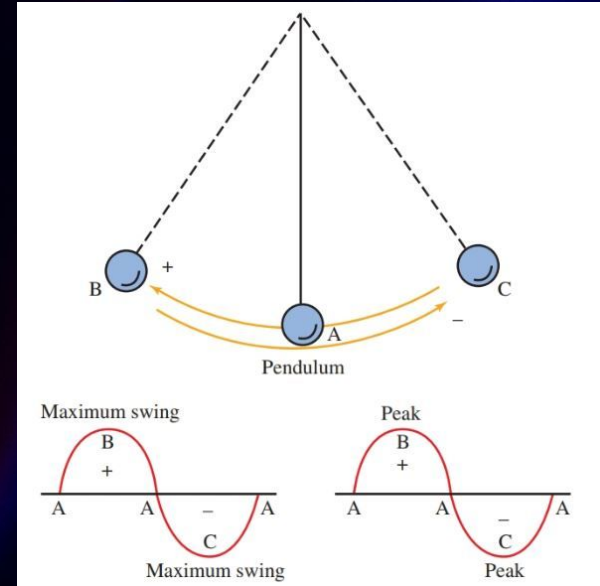
Spatial dynamics



swarms self-organise physically

+

Temporal dynamics



oscillators synchronise their internal rhythms in time

Kuramoto and Viscek models

Q1

How do the spatial-phase coupling parameter J and the phase coupling parameter K interact? Do they produce collective swarmalator states? Which types of emergent behaviours show up?

Q2

Does stronger phase syncing lead to faster and/or stronger spatial clustering? Does stronger spatial clustering accelerate phase syncing?

Q3

Do changes in the coupling parameters J and K create typical phase transitions between different macro states, do these transitions exhibit typical behaviour?

H1

5 different macro states are expected to form.

H2

Proximity increases the speed and amount of phase syncing

1. Phase syncing accelerates spatial clustering
2. Spatial clustering accelerates phase syncing

H3

We expect typical behavior e.g. power-law, static transitions, diverging susceptibility near critical points.

Article | [Open access](#) | Published: 15 November 2017

Oscillators that sync and swarm

[Kevin P. O'Keeffe](#), [Hyunsuk Hong](#) & [Steven H. Strogatz](#) 

Research Questions and Hypotheses

How do we model a 2D swarmalator?

- Each agent has:
 - a *spatial* position (x, y)
 - an *internal* phase (θ)
- Two-way coupling (J, K) :
 - phase affects motion
 - position affects phase
- Combines:
 - swarming models (e.g. Vicsek)
 - synchronization models (Kuramoto)

Coupled dynamics: space and phase

- Position update:
 - attraction-repulsion
 - attraction depends on phase similarity
- Phase update:
 - Kuramoto-like coupling
 - interaction strength depends on distance

$$\dot{\mathbf{x}}_i = \frac{1}{N} \sum_{\substack{j=1 \\ j \neq i}}^N \left[\frac{\mathbf{x}_j - \mathbf{x}_i}{|\mathbf{x}_j - \mathbf{x}_i|} (1 + J \cos(\theta_j - \theta_i)) - \frac{\mathbf{x}_j - \mathbf{x}_i}{|\mathbf{x}_j - \mathbf{x}_i|^2} \right],$$
$$\dot{\theta}_i = \frac{K}{N} \sum_{\substack{j=1 \\ j \neq i}}^N \frac{\sin(\theta_j - \theta_i)}{|\mathbf{x}_j - \mathbf{x}_i|}.$$

Key parameters: two way coupling

- J — phase $\rightarrow$ space coupling
 - How much phase similarity increases spatial attraction
- K — space $\rightarrow$ phase coupling
 - Strength of phase synchronization
 - $K > 0$: synchronization
 - $K < 0$: desynchronization
- Sweeping over (J, K) defines the phase diagram

Numerical implementation

- Agent-based simulation
- All-to-all interactions (N^2)
- Deterministic dynamics
- Explicit Euler integration
- Vectorized NumPy implementation

The background is an abstract, fluid pattern of deep blue and vibrant purple. The colors blend together in a way that suggests movement, like liquid or smoke. There are no sharp lines, only smooth, flowing gradients and subtle highlights that give the impression of depth and texture.

DEMO

How do we classify collective states?

Each simulation is classified using *four* diagnostics:

1. Is the system moving?
 - Mean velocity v
 - Mean angular velocity ω
2. Are phases globally synchronized?
 - Kuramoto order parameter R
3. Is phase correlated with position?
 - Correlation order parameter S
4. Is the phase structure continuous or discrete?
 - Number of phase groups
 - Spatial separation of those groups
 - Phase coherence within each group

This distinguishes all observed collective states.

Experimental setup

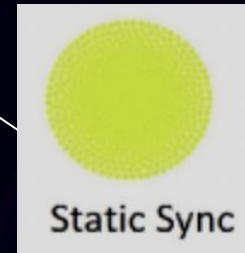
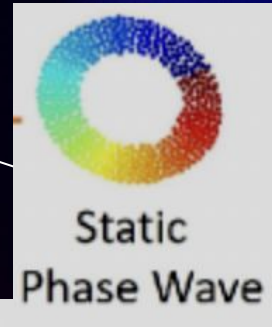
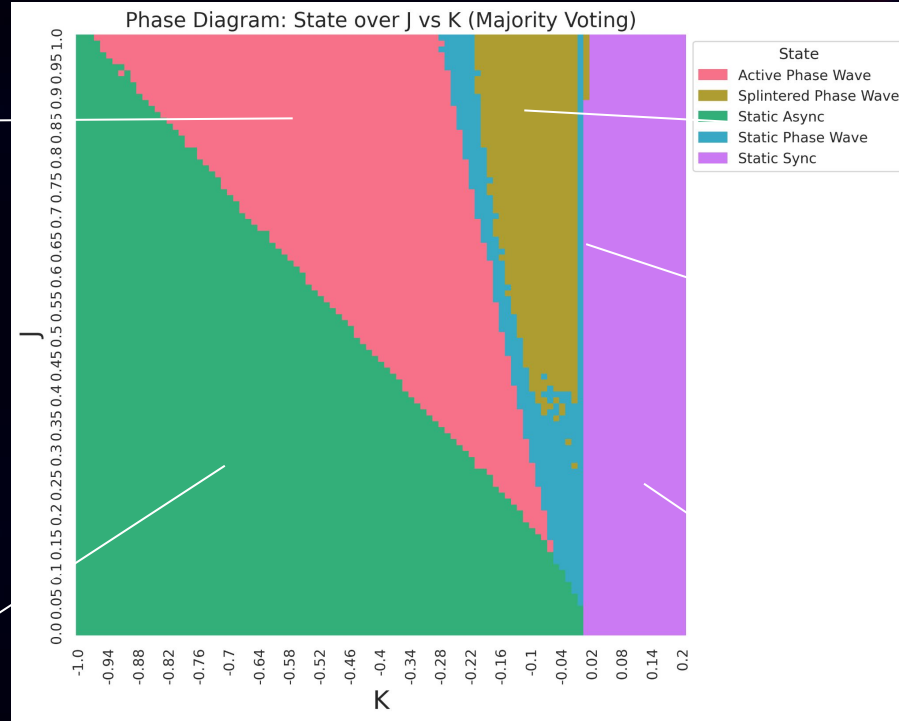
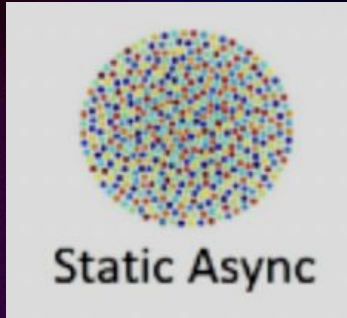
Swept Over:

- J and K
- N
- Seeds

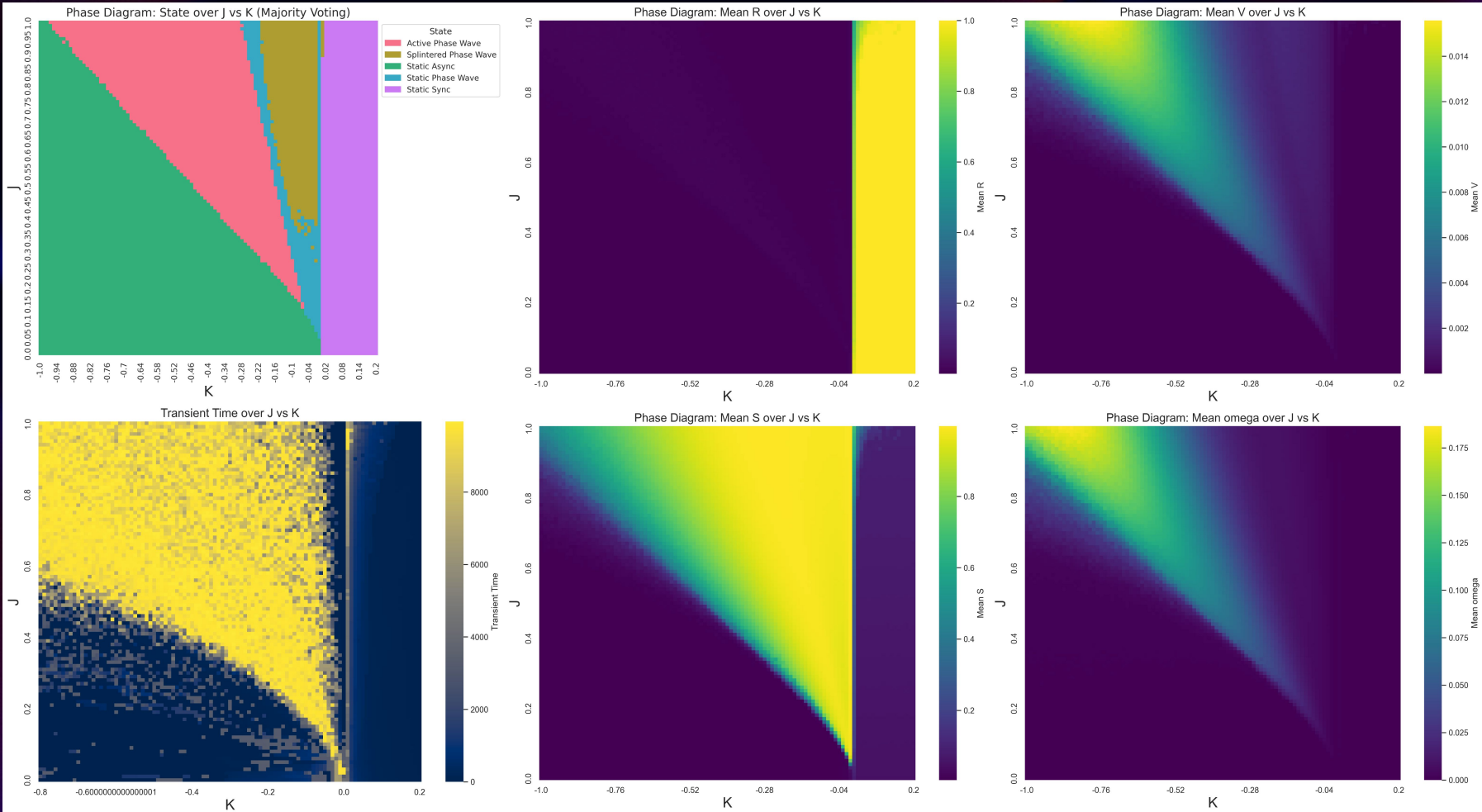
Graphs:

- Phase Diagrams
- Spearman Correlation
- Log-Log Plot for K critical point

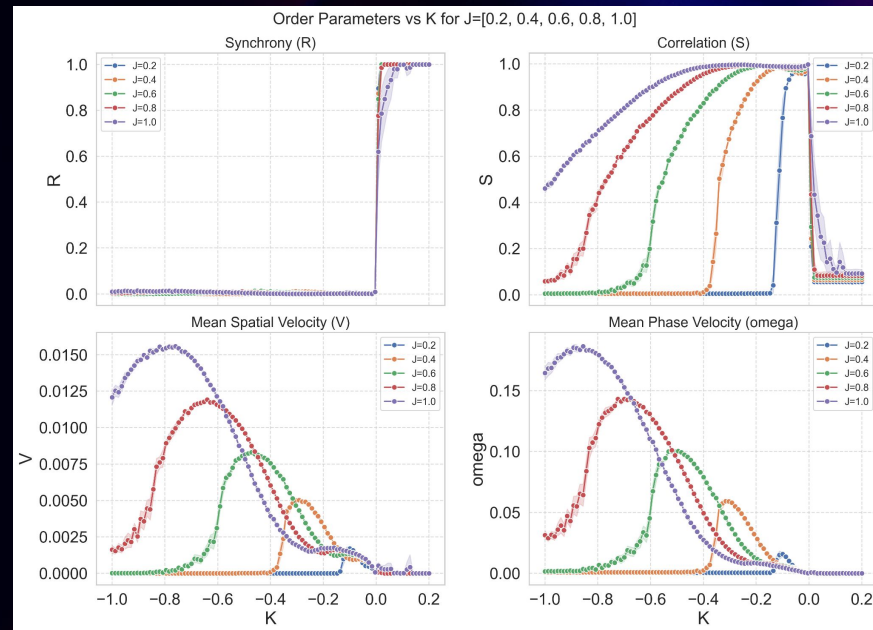
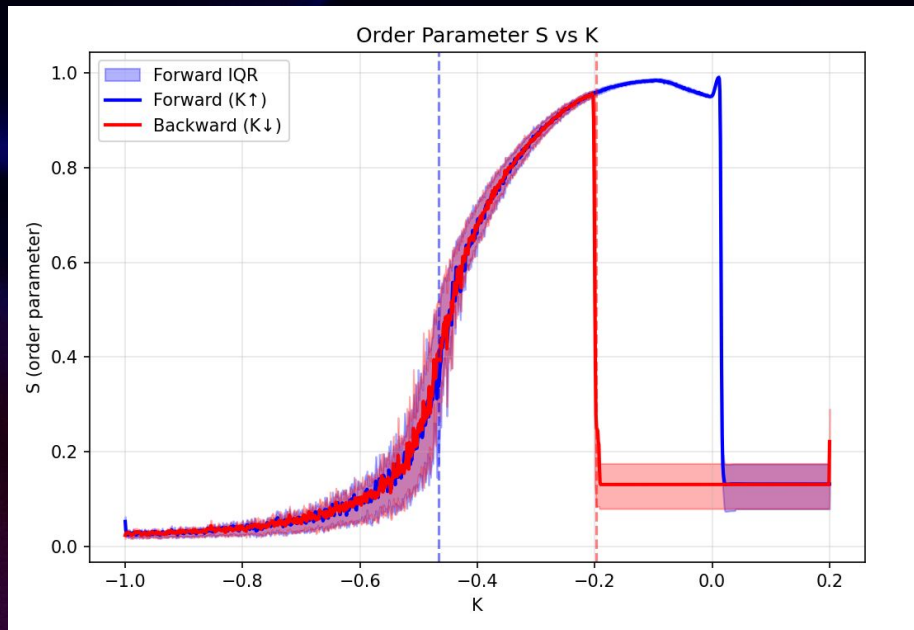
Phase Diagram



Phase Diagrams



Order Parameters & Hysteresis



95% confidence interval

where $J=0.5$

Finite Size Scaling

STATIC SYNC:

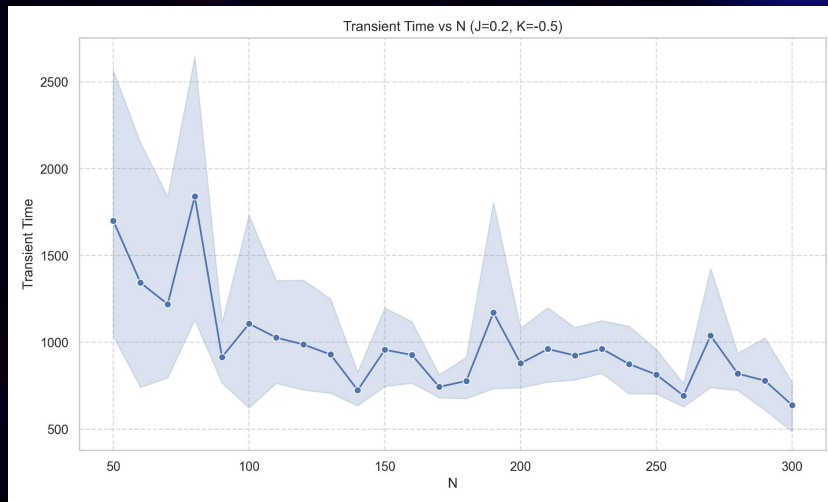
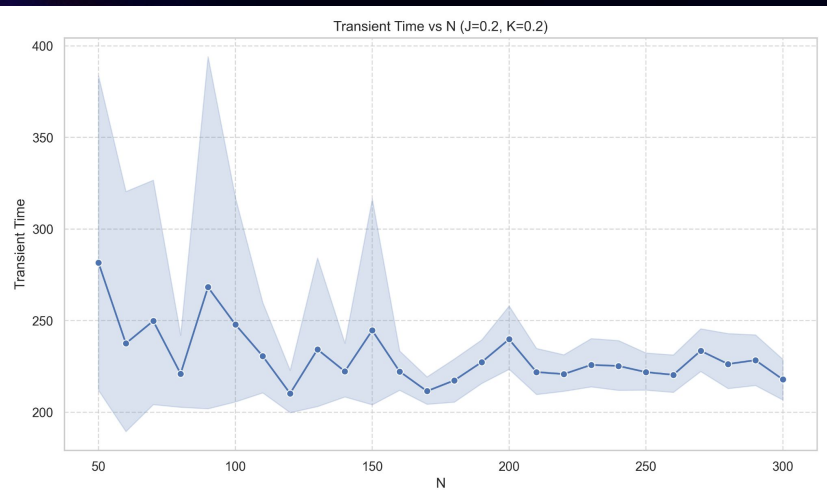
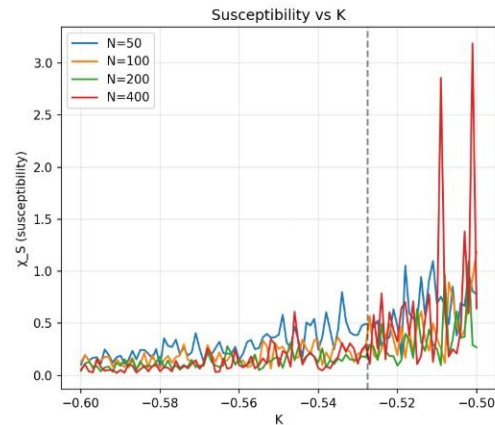
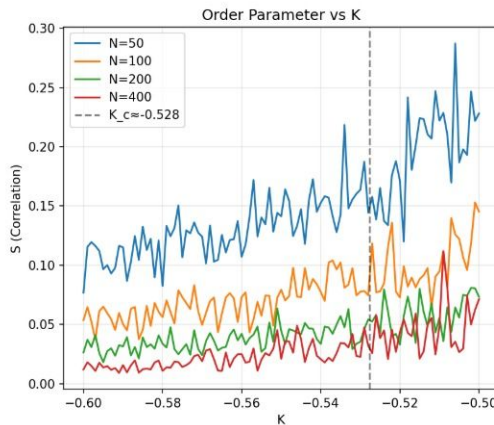
Spearman rho = -0.45784457478005863

p-value = 0.00841486751772789

STATIC ASYNC:

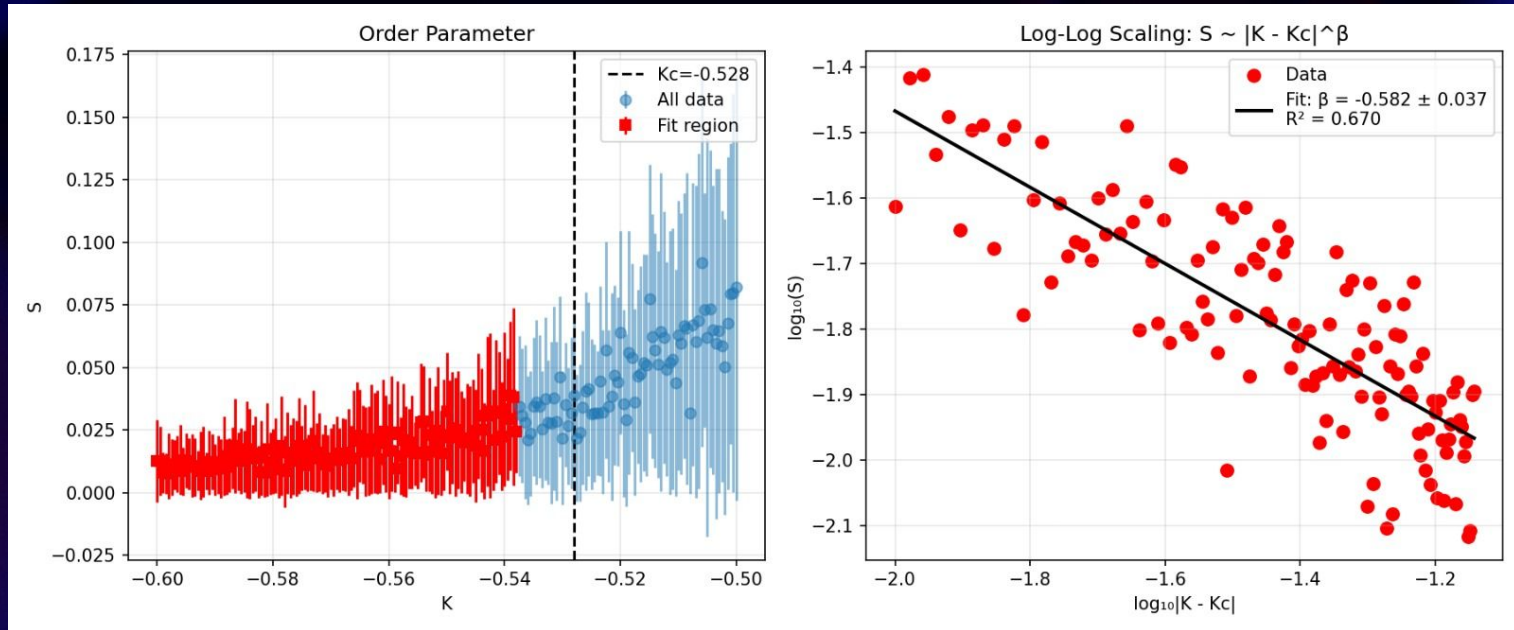
Spearman rho = -0.566502463054187

p-value = 0.0013563815557908923



Criticality Around Phase Transition

- Ran a small area around K_c for $J=0.5$
- Fit a power law relation for our order parameter



Limitations

O1 Computational Limitations

We look into only one phase transition. Many more phase transitions can be further explored. Run more experiments with higher number of agents N .

O2 Euler approximation

The original paper uses Heun's method (two-stage second-order Runge–Kutta methods), which leads to better approximation compared to ours.

O3 Model Extensions

Extensions are only visually examined and not analytically. The predator and chirality extensions can be further worked on.

O3 Inconclusive Evidence

Weak power law relationship.

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We expect typical behavior e.g. power-law, static transitions, diverging susceptibility near critical points.

A3

YES



A2

YES



A3

Yes, but NO - inconclusive



Conclusion

The background of the slide is an abstract, fluid pattern of deep blue and vibrant purple. The colors blend together in a way that creates a sense of movement, with soft, undulating lines and subtle gradients. The overall effect is a modern, artistic backdrop that contrasts with the simple, bold text.

Thank you!

Appendix

Interplay of sync and swarm: Theory and application of swarmalators

Oscillators that sync and swarm

Diverse behaviors in non-uniform chiral and non-chiral swarmalators | Nature Communications